
ECE228 Project Proposal: Trajectory Prediction of Marine Drifting Objects

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1 Problem Background and Motivation

Predicting the trajectories of marine drifting objects, such as ocean buoys, floating debris, and oil spill particles, is important for oceanography, environmental monitoring, and maritime search and rescue. Accurate drift prediction can help track marine pollution, support oil spill response, analyze ocean surface currents, and narrow down search regions after accidents.

A real-world motivation is the search for floating debris after aviation or shipping incidents. For example, after the disappearance of Malaysia Airlines Flight MH370, ocean drift modeling was used to estimate possible debris pathways and support search area analysis. Such cases show that reliable prediction of marine drifting motion can have significant practical value.

However, marine drifting trajectories are affected by complex factors such as ocean currents, wind, waves, and local environmental conditions. Simple physics-based models are interpretable but may be limited when the dynamics are nonlinear or when environmental data are noisy. In this project, we aim to use the NOAA Global Drifter Program 6-hour dataset¹ to predict future positions of drifting objects and compare physics-based baselines with machine learning and deep learning models.

2 Related Work

Prior work related to this project can be grouped into three main directions: physics-based drift modeling, Lagrangian particle-tracking frameworks, and data-driven trajectory prediction.

Physics-based drift modeling. Traditional marine drift prediction often models floating objects as Lagrangian particles transported by ocean currents, wind, and wave-induced effects. This approach has been used in real search and rescue scenarios, such as studies of possible MH370 debris drift paths [1, 2]. These models are interpretable, but their accuracy can be sensitive to uncertain environmental forcing, initial conditions, and object-specific drift behavior.

Particle-tracking frameworks. Numerical frameworks such as Parcels simulate ocean particle transport using gridded ocean circulation data [3]. They are useful for modeling passive particles such as plastics, biological material, or floating debris, but they depend strongly on the quality and resolution of the underlying ocean current fields. Our project instead focuses on learning motion patterns directly from observed drifter trajectories.

Data-driven trajectory prediction. Recent studies have applied machine learning and deep learning models, including LSTM, CNN-LSTM, ConvLSTM, and Transformer-based methods, to ocean trajectory prediction [4, 5]. These methods can capture nonlinear temporal patterns, but they may require sufficient training data and can be less interpretable than physics-based models.

Our project is situated between these directions. Using NOAA Global Drifter Program observations [6], we will compare a physics-informed advection baseline, classical machine learning models, and sequence-based deep learning models under the same trajectory prediction setting.

¹<https://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.nodc:AOML-GDP-6hr>

3 High-level Methodology

We propose to build a supervised learning pipeline for marine drifting object trajectory prediction. The main goal is to predict the future displacement of a drifting object based on its recent trajectory and velocity information. Instead of directly predicting absolute latitude and longitude, we will primarily predict displacement, since this representation is more stable across different ocean regions and better captures local motion patterns.

Given a drifter trajectory, we first construct training samples using a sliding time window. For each time step t , the input is a sequence of past observations:

$$X_t = \{(lat_{t-k}, lon_{t-k}, u_{t-k}, v_{t-k}), \dots, (lat_t, lon_t, u_t, v_t)\},$$

where (lat_t, lon_t) denotes the position of the drifter and (u_t, v_t) denotes the zonal and meridional velocity components. The prediction target is the future displacement after a fixed prediction horizon:

$$\Delta Y_t = (lat_{t+\Delta} - lat_t, lon_{t+\Delta} - lon_t).$$

The predicted future position can then be recovered by adding the predicted displacement to the current position.

Our methodology consists of three levels of models. First, we will implement simple physics-informed baselines. A persistence baseline assumes that the drifter remains at its current position, while an advection baseline assumes that the object moves according to its current velocity:

$$x_{t+\Delta} = x_t + v_t \Delta t.$$

These baselines provide interpretable reference points and allow us to evaluate whether machine learning models can improve over simple physical assumptions.

Second, we will implement classical machine learning models such as linear regression and random forest regression. For these models, the input trajectory window will be flattened into a feature vector. We may also include engineered features such as recent displacement, average velocity, velocity magnitude, direction, and temporal changes in velocity. These models are useful because they are relatively simple, efficient, and interpretable, while still being able to capture some nonlinear relationships in the data.

Third, we will explore sequence-based deep learning models. We plan to implement an LSTM model to capture temporal dependencies in the trajectory history. The LSTM will process the input sequence and use its hidden representation to predict the future displacement. We will also investigate a Transformer-based model, where positional encoding and self-attention are used to capture dependencies across different historical time steps. The Transformer may be beneficial when longer temporal context is important for predicting future drift motion.

We will compare all models using the same train, validation, and test splits. The main evaluation metrics will include mean absolute error, root mean square error, and geodesic distance error measured in kilometers. In addition, we will visualize predicted trajectories against ground-truth trajectories to qualitatively compare model behavior. Through this framework, we aim to understand whether data-driven models can improve marine trajectory prediction beyond simple physics-based baselines, and which modeling approach provides the best trade-off between accuracy, interpretability, and complexity.

References

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